

investigation_20260826

BOB-186 — Investigation: workable-items diff partial-read false-null

Revision: 1 **Last modified:** 2026-08-26T18:40:00Z **Scope:**
Investigation only. No source changed, no tracker DB or
Markdown mutated. **Label:** (T11/002-user-owned-downloads -
milos85vasic - ? - xhigh)

Verdict summary

Question	Answer
Does diff still false-green on a partial read?	NO — fixed in constitution 5979128 (2026-08-25 21:06), live in both shipped binaries
Is the DB↔Markdown pair actually in sync right now?	NO — 8 real divergences, independently confirmed
Would committing the DB now commit a false green?	NO — the gate correctly refuses; it would commit a <i>known-red</i> divergence

1. Reproduction — measured counts

Real corpus, gate-resolved binary (constitution/scripts/workable-items/bin/workable-items):

```
$ workable-items diff --db docs/workable_items.db --issues docs/Issues.md --fixed docs/Fixed.md
~ BOB-159 body differs (md=3366 bytes db=4917 bytes)
~ BOB-197 status: md="Queued" db="In progress"
~ BOB-197 body differs (md=2460 bytes db=4091 bytes)
~ BOB-198 body differs (md=2253 bytes db=3600 bytes)
~ BOB-199 body differs (md=3221 bytes db=6382 bytes)
- BOB-200 present in DB, absent in Markdown
```

```
- BOB-201 present in DB, absent in Markdown
- BOB-202 present in DB, absent in Markdown
diff: 8 difference(s) (compared 198 Markdown item(s) against 201
DB item(s); read docs/Issues.md, docs/Fixed.md)
EXIT=1
```

The BOB-186 shape (--issues without --fixed) no longer green-lights:

```
$ workable-items diff --db docs/workable_items.db --issues docs/
Issues.md
diff: refusing to report a DB-vs-Markdown verdict that would
silently exclude part of
the DB – the supplied Markdown path(s) do not account for 122
item(s) located in Fixed
(supply --fixed). ...
EXIT=1
```

With the opt-in --partial-scope, the verdict is scope-limited and never says “in sync”:

```
diff: PARTIAL SCOPE – 8 difference(s) within the tracker(s)
compared (compared 76
Markdown item(s); read .../Issues.md), and 122 of 201 DB item(s)
were NOT compared:
122 in Fixed (no --fixed supplied).
EXIT=1
```

2. Root cause (pre-fix sync.go, commit 5979128^)

Two independently-reasonable pieces combined into a false-null.

(a) Scope narrowing — pre-fix lines 852-863, the reverse “absent in Markdown” pass:

```
for _, d := range dbItems {
    if d.CurrentLocation == "Issues" && *issuesPath == "" {
        continue
    }
    if d.CurrentLocation == "Fixed" && *fixedPath == "" {
        continue
    }
    if !parsedSeen[itemKey(d)] {
        fmt.Printf("- %s present in DB, absent in Markdown\n",
            d.AtmID)
        differences++
    }
}
```

Its own comment states the intent: skip rows whose tracker the caller never supplied, “missing-in-Fixed lines would be their own class of false positive”. Correct as a false-positive guard — but the continue also removes those rows from differences entirely, and nothing compensated.

(b) Scope-unaware verdict — pre-fix lines 894-897:

```
if differences == 0 {
    fmt.Printf("diff: DB and Markdown are in sync (compared %d
        Markdown item(s) "+
        "against %d DB item(s); read %s)\n", len(parsed),
        len(dbItems), read)
    return exitOK
}
```

It prints `len(parsed)` (77 — only what was parsed) and `len(dbItems)` (184 — the whole DB), then decides purely on differences. **The verdict printed both mismatched numbers and never compared them.** A partially blind instrument returned the identical quiet green a genuinely-synced corpus returns — §11.4.201(6) FALSE-NULL.

Root cause in one line: *a false-positive guard silently narrowed comparison SCOPE while the verdict logic remained scope-unaware — no invariant tied covered-scope to verdict.* The §11.4.201(1) fix created its §11.4.201(6) mirror image.

3. The fix (live)

5979128 adds a scope-accounting guard (post-fix `sync.go:813–867`) keyed on **measured DB content**, not on flag presence, and makes the verdict SHAPE a function of the measured unaccounted set (`sync.go:1007–1032`). Comment at `sync.go:1007` is attributed verbatim to B0B-186. Regression guard:

`sync_diff_partial_scope_test.go`, 9 tests including `TestDiffCmd_IssuesOnlyDoesNotClaimSyncWhileFixedRowsUncompared`, `TestDiffCmd_PartialScopeOptInNeverClaimsSync`, `TestDiffCmd_IssuesOnlyIsCompleteWhenDBHasNoFixedRows` (the false-positive guard).

Both shipped binaries carry it:

Binary	mtime	sha256 (head)	guard string	PARTIAL SCOPE
constitution/ scripts/ workable- items/bin/ workable- items (gate-	Aug 25 21:50	298c3e66...	1	2

Binary	mtime	sha256 (head)	guard string	PARTIAL SCOPE
resolved first)				
constitution/ scripts/ workable- items/ workable- items	Aug 25 20:37	e5412ffd...	1	2

Grep control needle: known-present "DB and Markdown are in sync"
→ 1 hit; negative control ZZZ_NOT_PRESENT_NEEDLE_20260826 → 0 hits,
exit 1. Instrument proven seeing.

4. Control needles (§11.4.201(7)(b))

Needle	Method	Result
Negative control (false-positive guard)	regenerate MD from the COPY DB, diff	in sync (compared 201 ... against 201 ...), exit 0 — instrument is not stuck-on-FAIL, and 201-vs-201 is the counts-account-for- each-other property detected: ~ BOB-008
Needle A — Issues leg	flip BOB-008 status in the clean copy	status: md="In progress" db="Operator- blocked", exit 1 detected: - BOB-067
Needle B — Fixed leg (class-matched to BOB-186)	delete BOB-067 from the clean Fixed copy	present in DB, absent in Markdown (200 vs 201), exit 1 — the Fixed leg is genuinely READ, not merely “accounted for”

Each needle byte-verified applied before the run (before/after
grep counts differed).

5. Independent enumeration of the current divergence

Not trusting the tool — sqlite3 + grep on the real files:

```
db_issues=79  md_issues=76
-- in DB, NOT in Issues.md --    BOB-200  BOB-201  BOB-202
-- in Issues.md, NOT in DB --    (none)
db_fixed=122  md_fixed=122      (both directions empty)
```

BOB-197 independently confirmed: docs/Issues.md:1466 reads
Status: Queued; DB items.status = In progress.

Body sizes independently confirmed larger in DB for all four ~
body differs items (e.g. BOB-159 md-section 3267 B vs DB
length(body_md) 4905 B).

Direction of drift: the DB is AHEAD; docs/Issues.md is stale.
Regenerating from the DB produces a 193 436 B Issues.md vs the
committed 175 630 B. docs/Fixed.md is fully current (item-wise
byte-identical; the only delta is the §11.4.44 header, Rev 32 on
disk vs Rev 24 stored).

6. Blast radius for the imminent commit

- Worktree state: M docs/workable_items.db **only**. Issues.md / Fixed.md are clean at d7cf05d. So this is a DB-moved-ahead-of-Markdown state, not a race.
- core.hooksPath is UNSET (§11.4.234(B)) — automatic git hooks do NOT gate the commit.
- scripts/commit-push-all.sh:378,386 calls _docs_sync_seam_check unconditionally, which delegates to scripts/hooks/docs-sync-commit-seam.sh. Its CHECK 2 runs diff --issues --fixed and is **not** behind BOBA_SYNC_SKIP_CI (that flag gates only CHECK 4, line 327). CHECK 2 will FAIL → the commit is REFUSED.
- scripts/pre_build_verification.sh:585 invariant 17 also passes both paths → fail.
- **Residual risk:** a raw git commit bypasses all of it, because hooks are disconnected.

7. Remediation direction (NOT applied)

- **BOB-186 itself is already fixed** in code, binary, and test. Its tracker row is still Status: Queued in Issues. Remediation = close the item citing 5979128 + sync_diff_partial_scope_test.go; no code work is owed.

- **The current divergence** is a separate, live condition. Verified safe on copies: in-place sync db-to-md bumps Issues Rev 75→76 (monotone, §11.4.44 respected), leaves Fixed at 32 (content unchanged), and yields in sync (201 vs 201). Regenerating to a new path instead would regress the headers to the DB-stored 34/24 — use in-place.
- Choose the authoritative side first: DB-is-right → sync db-to-md; Markdown-is-right → sync md-to-db. Given BOB-200/201/202 exist only in the DB, DB-is-right is indicated.

8. Honest gaps

- UNKNOWN: whether the two shipped binaries are byte-for-byte reproducible from current HEAD source — not rebuilt (no .venv/build run authorised here). Both were proven to carry the guard and to behave identically on the real corpus, which is the property that matters for this defect.
- No claim is made about defect classes diff does not compare (revision headers, summary docs, Issues_Summary.md/Fixed_Summary.md); CHECK 3 of the commit seam and invariant 18 cover adjacent ground and were not exercised here.

Files read (no writes outside this directory)

constitution/scripts/workable-items/cmd/workable-items/
sync.go, .../sync_diff_partial_scope_test.go, scripts/
pre_build_verification.sh, scripts/hooks/docs-sync-commit-seam.sh,
scripts/commit-push-all.sh, docs/workable_items.db (read-only),
docs/Issues.md, docs/Fixed.md.

Post-run integrity check — real artifacts untouched:

ad8157f877a25f586d425e12b3ac596d383a794f900dcc5d3174f416b238894e
docs/workable_items.db
a8657a4280bc72f85fae59849acb33990a97500e353c1bc0c41807c9b9b8b3d2
docs/Issues.md
9354fe29b4d56f01969ba1642a3f8620a37466cc347e206a6870def2e8f57331
docs/Fixed.md